

Accepted Service Adaptation: Shared Policy Tables, Continuation Prices, and Certified Design

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Abstract. A service provider must respect promised continuations when deciding how much policy flexibility to install. We connect optimized information restrictions to a sufficient stochastic state by requiring one shared policy table to survive every successor problem. A bridge theorem represents the restricted optimum through inherited service tiers, remaining payment promises, and finite public memory. Recursively accumulated participation and restriction prices certify the optimized table through a root linear program; their release coefficient recovers the marginal value of flexibility. The representation is exact for the stated additive Markov institution, while finite stored plane families provide computable bounds without a query history vector. An explicit implementation-cost model converts information-value accounting into a net design decision. Certified witness widths govern price-cache refreshes. A second certificate covers declared parameter-dependent policies under jointly changing rewards, continuation coefficients, commitments, and switching-friction scale; it checks correlated interior feasibility rather than only parameter vertices. Exact examples expose invalid childwise table selection and endpoint-only feasibility. A scaling study reports full certificate costs, including frequent refreshes under strict tolerances. Earlier structural results and unfavorable learning evidence are retained. State sufficiency is separated from the dimension-dependent cost of certified approximation.

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1 Introduction

A service provider can find a more profitable operating policy and still be unable to offer it. A customer with an exit right evaluates the promised service and transfers remaining at the current review, not only the surplus forecast when the agreement was signed. An operational improvement must respect every continuation commitment, physical capacity, and the cost of changing service tiers.

The relevant comparison is with the best implementable restricted contract. A time-only schedule, a current-regime table, and a finite-memory rule use different organizational capabilities. Those capabilities cost money to install and maintain. Comparing adaptation with an arbitrary fixed protocol mixes the value of flexibility with poor selection of the original policy. We optimize both contracts and then optimize the net value of the implementable design.

The central difficulty is that a public-information restriction couples histories that a stochastic recursion would otherwise separate. A time-only schedule must remain the same schedule in every successor. Allowing each child to choose its own table removes the restriction and overstates its optimized value. Conversely, requiring the current public regime alone to summarize a contract can omit its inherited tier and outstanding payment promise. These are distinct failures of representation.

Our main contribution is a shared-table bridge connecting these two requirements. One finite-memory table is selected before uncertainty is resolved. Conditional on that table, the remaining payment promise and inherited tier support exact stochastic implementation. Participation prices accumulate along the nested continuations, while the common table preserves the global stationarity of the information restriction. A backward price construction then bounds the optimized restricted contract through a root table-design linear program. Its release coefficient is the discounted restriction rent; when the certificate is tight it supports the optimized value frontier. Thus the frontier, sufficient-state recursion, and comparator certificate are one theorem chain rather than independent uses of generic optimization tools.

The representation has explicit boundaries. It applies to the stated additive public-Markov institution and finite-memory pooling, not automatically to arbitrary signed or nonadditive contractual constraints. The recursion avoids a query history vector, but continuous promises, table dimension, and the number of stored planes still cost computation. Exact finite-horizon dual representation is an existence result, not a polynomial storage theorem. We give conditional approximation rates and distinguish them from a posteriori certificates.

The supporting structural results retain the optimized release frontier and comparator-adjusted continuation rents. Relaxing a continuation improves both optimized values but can reduce their difference; greater switching friction can increase the adaptation premium when the restriction itself forces switching. A declared cost of memory and release turns these value comparisons into an endogenous organizational choice. A rational example selects an interior release rather than either the original table or full flexibility.

Implementation needs both a feasible policy and a valid upper bound on its optimized comparator. Audited witness widths provide a computable price-cache refresh rule, including at face changes. For a declared parameter-dependent policy, correlated constraints cannot generally be verified at parameter vertices alone. We therefore give an interior-valid polynomial cell certificate and an exact positive adaptive example. These results retain the actual query commitments and friction scale without claiming that all cost geometry may change.

The numerical evidence separates exact structural checks, synthetic scaling, and inherited results. The scaling study charges coefficient construction, candidate generation, audits, cache evaluation, witness repair, and refresh, and compares them with an exact specialized restricted solve at the same tolerance target. Frequent refreshes under strict tolerances are reported rather than hidden. The earlier unfavorable learned-versus-classical record remains intact. Neither learning superiority nor a broad empirical speed advantage is needed for the shared-table theorem.

2 Relation to Dynamic Contracting and Optimization

Continuation states are established in dynamic contracting. Spear and Srivastava (1987) and Thomas and Worrall (1988) use recursive contractual promises in environments with incentive or enforcement constraints. In Operations Research, Chen et al. (2020) study optimal monitoring when unobserved effort changes the arrival of adverse events. Their formulation uses the agent’s promised continuation utility as a state, with promise keeping, incentive compatibility, and continued participation determining its evolution. Liang et al. (2023) study resource-allocation contracts that induce effort to reduce adverse events and derive implementable allocation structures using dynamic incentives and self-generating sets.

Our public-state institution does not solve those information problems. Its controls are contingent service tiers under externally fixed prices, and its binding constraints are continuation participation and physical or payment commitments. The question is which accepted deviations improve upon an already optimized information restriction. A *continuation price* here is a multiplier on a continuation participation constraint. It is not the customer’s promised utility. The promised payment in Section 6 is instead a primal state variable. This distinction prevents a dual representation from being mistaken for a new promised-utility construction.

The journal-local boundary also includes maintenance, capacity partnerships, and dynamic assignment. Tian et al. (2021) choose payments and potential termination to induce unobserved maintenance and repair effort. Liu et al. (2019) design renegotiated membership agreements and transfers under privately updated demand information. He (2023) derives deadline mechanisms for project assignment with a privately evolving cost type and limited capacity. Here demand and service are public, prices are fixed, and customer exit is a participation constraint rather than a privately informed report or an endogenous termination instrument. Switching friction enters the provider’s operating objective. The structural question is the value of relaxing a specified public-information restriction, and its implementation through an exact remaining-payment state; we do not claim a solution to those incentive-design problems.

Constrained Markov decision processes supply resource-state and occupation-measure methods (Altman, 1999). Strong convexity, normal cones, and sensitivity of convex values are standard (Boyd and Vandenberghe, 2004). Absolute-value switching is an instance of a generalized-lasso penalty (Tibshirani and Taylor, 2011). Explicit parametric quadratic programming supplies the piecewise-affine policy and piecewise-quadratic value structure used below (Bemporad et al., 2002). We do not claim any of these general principles as new. Our use of them distinguishes restriction rents from participation rents, identifies the continuation-slack event that terminates a profitable release, and accounts for reoptimization of the comparator. The transfer identity by itself establishes none of those value comparisons.

Robust set containment is also an established device (Ben-Tal and Nemirovski, 2000; Bertsimas and Sim, 2004). We use it only for certification and quantify its local loss. Numerical optimizers, including OSQP (Stellato et al., 2020), produce candidate decisions and prices. The evidence below accepts a decision only after a separate rational calculation has checked feasibility and its primal–dual bracket. The historical value-gradient and learned-price results remain in the accompanying research archive, with their assumptions and proofs intact; they are implementation results rather than the identity of this paper.

Nonanticipativity and policy aggregation already treat information bundles as constraints on scenario decisions (Rockafellar and Wets, 1991). Benders decomposition already separates shared design variables from conditional subproblems (Benders, 1962), and stochastic dual methods already propagate value approximations (Pereira and Pinto, 1991). The added conclusion here is not a new decomposition principle: it is that a *single shared table*, exact payment promises, and nested participation prices produce an exact restriction-value representation and a recursively priced optimized-comparator certificate. Without the table’s global stationarity, childwise optimization certifies the wrong class. Without the additive continuation structure, the displayed price recursion is not available. EC Section EC.16 compares every principal result with its closest established theorem class and states the essential assumptions.

3 A Service Agreement with Continuation Participation

3.1 Primitives and accepted continuations

At date $t = 0, \dots, T - 1$, a provider observes a public regime z_t and an inherited tier q_t . It chooses stock S_t and a tier $x_t \in [0, 1]$ before demand is realized. Demand D_z is exogenous, nonnegative, and has finite mean. Regime transitions do not depend on the policy. The next inherited tier equals the chosen tier. Premium $r_z x$, indemnity $\kappa x(D_z - S)^+$, and physical cost parameters are fixed in the agreement. There is no private information. With $f_z(S) = \mathbb{E} \min(S, D_z)$ and $L_z(S) = \mathbb{E}(D_z - S)^+$, the customer’s expected stage utility is

$$u(z, S, x) = \nu f_z(S) - a(z, S)x, \quad a(z, S) = r_z - \kappa L_z(S), \quad \nu > 0. \quad (1)$$

Maintenance and switching costs are resources expended by the provider, not transfers received by the customer.

An operational interpretation is a service-level agreement or maintenance subscription with verifiable fulfillment and periodic cancellation rights, rather than an unrestricted mechanism-design problem. The customer may leave *before implementation at every review*. Remaining contractual utility must exceed the publicly specified outside continuation $\underline{U}(h_t)$ at every positive-probability history h_t . The provider commits to its announced contingent protocol; the exit right does not let it retrospectively change prices or tiers. We consider contracts intended to retain the customer at all modeled reviews. A model in which termination is itself an optimized action would need an additional termination decision.

Stock can be contractually specified. It can also be identified from service and cost commitments: if s_z uniquely minimizes $C_z(S) - \xi f_z(S)$, $\xi > 0$, then $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ at every reachable review. The nonnegative expected sum of the minimization gaps must be zero. EC Section EC.1 gives the

boundary, vector, and relaxed-commitment statements. Thus the tier reduction need not assume away a stock decision that the accepted physical commitments already determine. Without such identification, stock must remain in the accepted optimization.

On a finite exogenous tree, let $w_n = \beta^{t(n)} \Pr(n) > 0$, $c_n = w_n a(z_n, s_{z_n})$, and $i \succeq n$ mean that i is in the subtree rooted at n . Multiplying conditional participation by w_n gives

$$\sum_{i \succeq n} c_i x_i \leq b_n, \quad b_n = \sum_{i \succeq n} w_i v f_{z_i}(s_{z_i}) - w_n \underline{U}(n). \quad (2)$$

This derivation imposes every continuation constraint directly. Ex ante participation alone is the special case retaining only the root row. It does not imply the nonroot inequalities. A fixed root payment, when stipulated in the agreement, gives $\sum_i c_i x_i = b_0$; it is a separate payment commitment, not an equality inferred from participation. General formulations below also allow a root inequality.

3.2 Optimized restrictions and switching

Write the accepted set, including the tier box, as

$$X_b = \{x \in C : Ax \leq b, Ex = e\}, \quad C = [\ell, u], \quad W_\lambda(x) = p^\top x - \frac{1}{2}x^\top Qx - \lambda T(x), \quad (3)$$

where $Q \succ 0$, $\lambda \geq 0$, and

$$T(x) = \sum_{a=1}^m \kappa_a |(Dx - h)_a|, \quad \kappa_a > 0.$$

The linear reward p includes net receipts and the linear part of convex operating costs, for example a quadratic cost around a service-specific efficient tier. It need not be proportional to the payment vector c . The general quadratic model and its synthetic examples are not asserted to be estimates of the stock institution above.

Rows of A include (2) and any other retained capacity constraints. E contains fixed payment or physical equalities. D can include temporal and cross-service differences; h incorporates an installed initial tier. The vector-service case uses the same stacked notation, and signed payment coefficients are allowed in (3). Only the specialized transfer lemma requires positive scalar coefficients. All accepted sets used below are nonempty and compact. Strong concavity makes each optimum unique.

A contractual restriction is a predeclared equality $Mx = m_0$, and

$$Y_b = X_b \cap \{Mx = m_0\}, \quad K_\lambda(b) = \max_{x \in Y_b} W_\lambda(x), \quad J_\lambda(b) = \max_{x \in X_b} W_\lambda(x), \quad \Delta_\lambda(b) = J_\lambda(b) - K_\lambda(b). \quad (4)$$

The matrices and commitments are the same on both sides. Pooling decisions at histories with the same permitted information gives one important M . The experiments also pool *amendments* $x - z^{\text{legacy}}$; then $m_0 = Mz^{\text{legacy}}$, which is generally different from pooling actual tiers. We keep that distinction explicit.

3.3 Continuation prices and relative transfers

For a comparator $z \in X_b$, put $g = p - Qz$. Convex optimality gives the following retained foundation.

Proposition 3.1 (Accepted balance). *The comparator z is optimal over X_b if and only if there exist $\mu \geq 0$, unrestricted η , $v \in N_C(z)$, and switching tensions s such that*

$$g = A^\top \mu + E^\top \eta + v + D^\top s, \quad \mu_i(A_i z - b_i) = 0, \quad (5)$$

with $s_a = \lambda \kappa_a \text{sign}(Dz - h)_a$ on nonzero differences and $|s_a| \leq \lambda \kappa_a$ on zero differences. If it is not optimal, a feasible sufficiently short direction has strictly positive directional gain. For optimality over Y_b , add $M^\top \xi$ to (5); ξ prices the restriction rather than a customer participation constraint.

Proof. The objective loss is the sum of a differentiable convex quadratic and a finite convex absolute-value function. Its subgradient plus the polyhedral normal cone characterizes a minimum. Polyhedral normals give the displayed multipliers, without a strict-feasibility assumption. The alternative follows from the directional derivative test for a convex function on a polyhedron. $\square$

In the scalar tree specialization, assume $c_n > 0$ and a fixed root payment. For any feasible z , define its continuation slack $\sigma_n = b_n - \sum_{i \geq n} c_i z_i$ at nonroot nodes. Let B have column n equal to -1 at row n and $c_n/c_{\text{pa}(n)}$ at its parent, with zeros elsewhere.

Lemma 3.2 (Benchmark-relative transfer coordinates). *The correspondence*

$$x = z + By, \quad y_n = -c_n^{-1} \sum_{i \geq n} c_i (x_i - z_i) \quad (6)$$

is a bijection between X_b and the relative transfers satisfying $y_n \geq -\sigma_n/c_n$, the original tier box, and every additional inequality or equality after substitution. The restriction $Mx = Mz$ becomes $MBy = 0$. Both directions of the map take linear work in the number of tree nodes.

Subtree telescoping proves the lemma; EC Section EC.2 gives the inverse explicitly. Negative transfers consume pre-existing continuation slack and cannot be discarded. The lemma is a representation, not a theorem about how much an optimized restriction costs. We next use the restriction prices and the slack in a value theorem.

4 The Value Frontier of Releasing a Restriction

Fix b, λ and the optimized comparator $z = \arg \max_{Y_b} W_\lambda$. Let $\omega \in \mathbb{R}_{++}^k$ set the units of permitted deviations from the restriction. Define

$$V(t) = \max\{W_\lambda(x) : x \in X_b, |Mx - m_0| \leq t\omega\}, \quad t \geq 0. \quad (7)$$

Thus $V(0) = K_\lambda(b)$. The release does not relax participation: it permits previously pooled decisions to differ while preserving every accepted constraint. The full value is reached for finite t , because X_b is compact. Scaling a row of M requires the same scaling of its ω ; otherwise it changes the economic release unit.

4.1 An exact release segment and its first bottleneck

To state the explicit segment, fix a regular active cell at z . Collect permanent equalities, active accepted inequalities and tier bounds, and zero switching differences into $Rx = r$. Retain the signs of all nonzero switching differences and absorb their linear cost into $\bar{p}$. Let R and $[R^\top, M^\top]^\top$ have full row rank. Suppose the active accepted-inequality multipliers are strictly positive, zero-edge tensions are strictly within their capacities, inactive constraints and nonzero switching signs have positive clearance, and the restriction multiplier ξ has no zero coordinate. These are local regularity conditions, not assumptions of the global frontier in the theorem. Rows with zero switching coefficient need not be included as kink constraints.

Choose any matrix Z whose columns form a basis for $\ker R$, and set

$$H = Z^\top QZ, \quad S = [MZH^{-1}Z^\top M^\top]^{-1}, \quad P = ZH^{-1}Z^\top M^\top S, \quad d = Pv, \quad v_j = \omega_j \text{sign } \xi_j. \quad (8)$$

These objects are independent of the choice of basis. S is the marginal curvature of releasing the restriction after the currently binding accepted constraints have been enforced.

Theorem 4.1 (Restriction-release frontier). *For (3)–(7), V is continuous, nondecreasing, concave, and piecewise quadratic. Its unique tier policy is continuous and piecewise affine. At almost every $t > 0$,*

$$V'(t) = \omega^\top (\xi^+(t) + \xi^-(t)), \quad V(t) - V(0) = \int_0^t \omega^\top (\xi^+(s) + \xi^-(s)) ds, \quad (9)$$

where $\xi^\pm$ are optimal multipliers on the two release bounds. At differentiability points the displayed directional price is independent of the choice of optimal multipliers.

Under the regular-cell conditions, let t_1 be the first nonnegative time at which an inactive accepted constraint or tier bound binds, a nonzero switching difference reaches zero, an old active-inequality multiplier reaches zero, a zero-edge tension reaches a capacity, or a component of $\xi - tSv$ reaches zero in its original signed direction. Then $t_1 > 0$ and, for $0 \leq t \leq t_1$,

$$x(t) = z + td, \quad V(t) - K_\lambda(b) = t \sum_j \omega_j |\xi_j| - \frac{1}{2} t^2 v^\top S v. \quad (10)$$

Each event time is an explicit linear ratio obtained from the primal and dual paths. In particular, a currently slack continuation row can be the first bottleneck only at

$$t_n = \frac{\sigma_n}{\sum_{i \geq n} c_i d_i} \quad \text{when } \sum_{i \geq n} c_i d_i > 0; \quad (11)$$

otherwise that row consumes no slack along this segment. In transfer coordinates this is exactly the time at which its lower bound $y_n \geq -\sigma_n/c_n$ is reached.

Proof. Mixing policies feasible at two release levels proves concavity; nesting proves monotonicity. Introduce epigraphs for the absolute values. For fixed active inequalities and switching signs the KKT equations are linear in the parameter, so the unique tier solution is affine and its value is quadratic on that cell. There are finitely many active patterns. Dependent equations may be removed and a finite refinement made; no

nondegeneracy assumption is needed for this global piecewise description. Compactness and uniqueness give continuity. Optimal multipliers support the concave value and give (9); integration is valid for this continuous piecewise-quadratic function.

For the explicit segment, a prescribed restriction displacement δ has the cell optimum $x = z + P\delta$. Solving the equality-constrained quadratic and using $RP = 0$, $MP = I$, and $P^\top QP = S$ gives the gain $\xi^\top \delta - \delta^\top S\delta/2$. Its maximum over $|\delta| \leq t\omega$ is at $\delta = t\nu$ while $\xi - tS\nu$ retains its signed direction. The old active multipliers and kink tensions also evolve affinely. Until one of the listed events, their signs and capacities, the remaining primal inequalities, and the nonzero switching signs all remain valid. They therefore supply the *full* KKT certificate, not just a solution restricted to the cell. Continuity includes the first event. Substitution into a subtree inequality gives (11). The detailed dual path and degeneracy qualifications are in EC Section EC.3. $\square$

Write $a = \sum_j \omega_j |\xi_j|$ and $q = \nu^\top S\nu > 0$. For a desired gain $g \in [0, V(t_1) - K]$, the smallest release on this initial segment is

$$t(g) = \frac{a - \sqrt{a^2 - 2qg}}{q}. \quad (12)$$

No accepted improvement can attain this gain with a smaller release, because (10) is the optimized value, not the gain of a selected direction. At later events one changes the active cell and applies the same equality-constrained calculation. The number of cells need not be polynomial; the theorem is an exact structural path, not a universal complexity claim. If a regularity condition fails, (9) and the global frontier remain valid, but the signed-corner formula is replaced by the appropriate small release quadratic program or a degenerate active-set refinement.

4.2 A complete three-node frontier

Consider a root and two children, $c_i = 1$, tiers in $[0, 1]$, root payment $x_0 + x_1 + x_2 = 3/2$, and child caps $x_1, x_2 \leq 3/4$. Let

$$W(x) = x_1 - \frac{1}{2}(x_0^2 + x_1^2 + x_2^2), \quad Y = \{x \in X : x_1 = x_2\}.$$

The optimized restriction is $z = (1/6, 2/3, 2/3)$ with value $5/24$. Releasing $|x_1 - x_2| \leq t$ gives

$$V(t) - K = \begin{cases} t/2 - t^2/4, & 0 \leq t \leq 1/6, \\ -1/48 + 3t/4 - t^2, & 1/6 \leq t \leq 3/8, \\ 23/192, & t \geq 3/8. \end{cases} \quad (13)$$

The first policy segment is $(1/6, 2/3 + t/2, 2/3 - t/2)$. Child 1 has slack $1/12$, consumes it at rate $1/2$, and therefore binds at $t = 1/6$. The second segment is $(t, 3/4, 3/4 - t)$. At $t = 3/8$, the release price becomes zero and the full optimum $(3/8, 3/4, 3/8)$ is reached. Its relative transfer is $(-1/12, 7/24)$; the negative coordinate uses exactly the initially available child-1 slack. Thus the continuation slack determines an actual change in the optimal policy path, rather than only the domain of a coordinate map. EC Section EC.4 supplies exact primal and dual witnesses for both segments.

5 Capacity, Friction, and Optimized Comparator Rents

5.1 Relaxing a continuation can reduce the adaptation premium

Let $b(t) = b^0 + td$, with $d \geq 0$, and suppose $Y_{b(t)}$ is nonempty on the interval considered. Both optima in (4) are reoptimized for each t .

Theorem 5.1 (Comparator-adjusted continuation rents). *The full and restricted values are continuous, nondecreasing, concave, and piecewise quadratic along the budget path. If $\mu_X(t)$ and $\mu_Y(t)$ are optimal accepted-inequality multipliers, then, at almost every t ,*

$$\frac{d}{dt}\Delta_\lambda(b(t)) = d^\top(\mu_X(t) - \mu_Y(t)), \quad \Delta_\lambda(b(v)) - \Delta_\lambda(b(u)) = \int_u^v d^\top(\mu_X - \mu_Y) dt. \quad (14)$$

Consequently, the adaptation premium is nondecreasing along the path if and only if the rent difference is nonnegative almost everywhere. At a degenerate point, the right directional value derivative along d is the minimum of $d^\top \mu$ over that problem's optimal multiplier set; one must compute the two directional derivatives separately, rather than subtract arbitrary selected multipliers.

Proof. Mixing and nesting give the first two value properties. Fixed-matrix parametric quadratic programming gives the piecewise description as in Theorem 4.1. Weak duality makes each accepted multiplier a supporting price for the budget. At differentiability points all such directional prices coincide with the derivative. Subtract and integrate the two value derivatives. A continuous piecewise-quadratic function is nondecreasing precisely when its derivative is nonnegative almost everywhere. The one-sided rule follows by taking the directional derivative of the infimum of affine-in-budget dual objectives; EC Section EC.5 supplies the polyhedral qualification. $\square$

An exact example makes the distinction material. Keep the preceding three-node objective and root payment, but set $x_1 \leq b$, $x_2 \leq 3/4$, with $b \in [1/4, 2/3]$. The time-only restriction remains $x_1 = x_2$. The full and restricted optima are

$$x^X(b) = ((3/2 - b)/2, b, (3/2 - b)/2), \quad x^Y(b) = (3/2 - 2b, b, b),$$

with cap prices $\mu_X = 7/4 - 3b/2$ and $\mu_Y = 4 - 6b$. Both values increase, but

$$\Delta(b) = \frac{9}{16}(2b - 1)^2, \quad \Delta'(b) = \frac{9}{4}(2b - 1). \quad (15)$$

The premium falls to zero as b rises to $1/2$, then rises again. A claim that more participation slack necessarily makes adaptation more valuable would get the sign wrong on half this interval. The relevant scarcity measure is the difference of the optimized contractual rents.

5.2 Friction thresholds when the restriction is reoptimized

For fixed X, Y , let x_λ and z_λ be their optima. A fixed-comparator balance test is not enough when z_λ changes with friction.

Proposition 5.2 (Optimized-comparator friction path). *The adaptation premium is continuously differentiable for $\lambda > 0$, with a right derivative at zero, and*

$$\Delta'(\lambda) = T(z_\lambda) - T(x_\lambda). \quad (16)$$

On every bounded friction interval its zero set is a finite union of closed intervals and isolated points. It can be computed from the restricted parametric policy path without solving a full-class optimization at every friction: refine that path at changes of its accepted active rows and switching signs, and on each resulting cell impose (5) at the affine z_λ . This is a linear feasibility problem jointly in λ and the full-class prices; its projection on λ is an interval, possibly empty or a point. The union of these projections is exactly the zero set. This is a structural reduction, not a general complexity improvement. There is no general upward-ray conclusion for arbitrary restrictions.

Proof. Uniqueness and compactness imply continuous optimizers. The envelope argument gives $J'(\lambda) = -T(x_\lambda)$ and $K'(\lambda) = -T(z_\lambda)$, including active-set changes. Both paths are piecewise affine. On a refined cell, every sign in (5) is fixed or zero, z_λ is affine, and tension bounds and stationarity are linear in the displayed variables. The full balance test is necessary and sufficient. A projection of a polyhedron onto a line is an interval. Finitely many cells establish the assertion; the method does not bound their number polynomially. $\square$

For example, use the root payment $3/2$, the box and child caps above, $p = (1/2, 3/4, 1/4)$, $Q = I$, and $T(x) = |x_1 - x_0| + |x_2 - x_0|$. A restriction $x_1 - x_2 = 1/2$ has optimized policy $z = p$ for every λ : its switching cost is already the smallest possible under that restriction. The full optimum is $(1/2, 3/4 - \lambda, 1/4 + \lambda)$ until $\lambda = 1/4$, and is constant thereafter. Therefore

$$\Delta(\lambda) = \begin{cases} \lambda^2, & 0 \leq \lambda \leq 1/4, \\ \lambda/2 - 1/16, & \lambda \geq 1/4. \end{cases} \quad (17)$$

More friction increases the value of removing a restriction that forces switching. This does not contradict constant-comparator critical-friction results: their restriction and zero-switching benchmark are different.

5.3 Which information expansion creates the value?

Let $Y_0 \subseteq \dots \subseteq Y_k = X$ be nonempty predeclared classes and $K_j = \max_{Y_j} W$. Then

$$J - K_0 = \sum_{j=1}^k (K_j - K_{j-1}), \quad K_j - K_{j-1} \geq 0. \quad (18)$$

The identity is elementary; its economic use requires each K_j to be optimized under the *same* accepted constraints. A release path between adjacent equality-defined classes prices that particular increment through Theorem 4.1. The ordering matters when information features interact; the increments are not automatically Shapley values or unique causal attributions.

The fixed class describes an unchanging amendment or tier. The time-only class describes a preannounced schedule. The current-regime class requires public regime measurement and a date–regime decision table. A two-period-memory class additionally requires retaining the preceding regime. The full-history class has no such information pooling. These are operational restrictions, not an assertion that time-only is the universal status quo. Current regime alone can be insufficient even when exogenous primitives are Markov, because the inherited tier and the outstanding promise are endogenous state variables. The next section establishes the corresponding sufficient-state formulation.

6 Promised-Payment Recursion and Horizon Complexity

Suppose the regime has a finite public Markov transition matrix P_t , the one-period net payment is $a_t(z)x$, and the reward is $g_t(z, q, x)$. Suppose further that each conditional payment cap $\bar{b}_t(z)$ and the feasible actions depend only on the displayed state and date. A general history-dependent outside option does not satisfy this assumption merely because demand is Markov.

Let $\mathcal{V}_t(z, q, b)$ denote the maximum conditional reward from date t with an *exact* remaining discounted payment promise b . Put $\mathcal{V}_T(z, q, 0) = 0$ and $\mathcal{V}_T(z, q, b) = -\infty$ for $b \neq 0$. For $b \leq \bar{b}_t(z)$, define

$$\begin{aligned} \mathcal{V}_t(z, q, b) = \max_{x, \{b_{z'}\}} & \quad g_t(z, q, x) + \beta \sum_{z'} P_t(z, z') \mathcal{V}_{t+1}(z', x, b_{z'}) \\ \text{subject to} & \quad a_t(z)x + \beta \sum_{z'} P_t(z, z') b_{z'} = b, \\ & \quad b_{z'} \leq \bar{b}_{t+1}(z') \quad \text{for positive-probability children.} \end{aligned} \tag{19}$$

Outside this cap, set the value to $-\infty$. A zero-probability successor imposes no obligation. For a root upper budget rather than an equality, maximize the root value also over permitted b .

Theorem 6.1 (Exact sufficient-state implementation). *Under the stated Markov sufficiency and compact-attainment conditions, (19) has the same value as the full accepted history-tree problem with its specified root promise. There exists an optimal deterministic contract implemented using only (t, z, q, b) . Conversely, a feasible recursion policy induces all tree continuation constraints. With several additive commitments, the result holds with a vector of remaining commitments. A history-dependent cap, nonadditive global resource, or omitted physical state must be included in the state or treated on the full tree; otherwise the claimed equivalence need not hold.*

Proof. For any history policy, take its conditional remaining payments in each child subtree as the child promises. They satisfy the equality and all child caps, and induction bounds its conditional reward by the right side. Conversely, combine an attained current maximizer with child maximizers at the allocated promises. Induction gives every child obligation, while the displayed equality gives the parent’s exact promise. This proves equality and constructs a state-based policy. The vector case applies the same induction to each additive commitment. $\square$

Conditional normalization is essential. Multiplying the conditional budget at node n by w_n recovers

(2). Within (19), the child weights are $\beta P_t(z, z')$, not unconditional probabilities multiplied a second time. Promises are decision variables allocated across next-period regimes; they are not forecasts fixed independently of the current decision.

A branching factor B and T decision dates can create $N = (B^T - 1)/(B - 1)$ history nodes. The linear transfer map is linear in N , not in T . The recursion avoids materializing those histories when the displayed state is sufficient, but its promise state is continuous in the continuous-tier model. A grid with G promise points, Q_0 inherited-tier points, S_0 regimes, and k commitment coordinates has order $TS_0Q_0G^k$ value entries; choosing child allocations is an additional cost and can grow exponentially in the number of successors. Such a grid is an approximation unless the contract institution itself has the stated finite actions and attainable promises. Our exact recursion checks use the latter finite-action interpretation, and exhaustively compare every attainable root promise with the full history enumeration. No interpolation theorem or unrestricted polynomial-horizon guarantee is inferred from them.

6.1 Continuous promises with executable error certificates

The sufficient-state statement is not yet an approximation guarantee. Interpolating a value table can violate an exact promise or admit a customer continuation that the contract excludes. We next construct both a feasible policy and a value bound without rounding the promise to a grid point.

Specialize to $x, q \in [0, 1]$, $a_t(z) > 0$, and

$$g_t(z, q, x) = r_t(z)x - \frac{1}{2}d_t(z)x^2 - \ell_t(z)|x - q|, \quad d_t(z) > 0, \quad \ell_t(z) \geq 0. \quad (20)$$

Only the Markov continuation caps and the exact payment commitment are imposed in this specialization; additional nonlocal constraints require additional state. Write $I_t(z) = [L_t(z), U_t(z)]$ for feasible remaining promises. With $L_T = U_T = 0$, their endpoints satisfy

$$L_t(z) = \beta \sum_j P_t(z, j) L_{t+1}(j), \quad U_t(z) = \min \left\{ \bar{b}_t(z), a_t(z) + \beta \sum_j P_t(z, j) U_{t+1}(j) \right\}. \quad (21)$$

A state is infeasible if a positive-probability successor is infeasible or the displayed lower endpoint exceeds the upper endpoint. All statements below are on feasible states; $\mathcal{D}_t(z) = [0, 1] \times I_t(z)$ includes the boundary.

At each date and regime, retain finitely many anchors (q_i, b_i, l_i) and affine upper planes $H_k(q, b) = A_k q + B_k b + C_k$. An anchor includes a feasible current action and mixtures of child anchors with the exact inherited tier and remaining promise, not just a scalar value. The anchor coordinates span $\mathcal{D}_t(z)$. Define

$$\begin{aligned} \underline{V}_t(z, q, b) &= \max_{\alpha \geq 0} \left\{ \sum_i \alpha_i l_i : \sum_i \alpha_i = 1, \sum_i \alpha_i (q_i, b_i) = (q, b) \right\}, \\ \bar{V}_t(z, q, b) &= \min_k H_k(q, b). \end{aligned} \quad (22)$$

To generate a parent plane, choose any convex mixture of child planes for each successor j , with coefficients

(A_j, B_j, C_j) . Suppress the parent date and regime in a, r, d, ℓ, P . For any $s \in [-\ell, \ell]$ and $\eta \in \mathbb{R}$, put

$$\begin{aligned} H(q, b) &= sq + \eta b + C, \\ C &= \psi_d \left(r - s - a\eta + \beta \sum_j P_j A_j \right) + \beta \sum_j P_j \left[C_j + \max_{v \in I_{t+1}(j)} (B_j - \eta)v \right], \\ \psi_d(v) &= \max_{0 \leq x \leq 1} \{vx - dx^2/2\}. \end{aligned} \quad (23)$$

The maximizer in ψ_d is $\min\{1, \max\{0, v/d\}\}$. Thus a proposed plane has a checkable global constant even when its prices were obtained only approximately.

Theorem 6.2 (Promise-preserving continuous-state envelopes). *Under (20) and the stated Markov assumptions, the interval recursion (21) is exact and $\mathcal{V}_t(z, \cdot, \cdot)$ is concave on $\mathcal{D}_t(z)$. Start with zero terminal values and construct anchors and planes backward as above. If every anchor reward is a lower bound for its displayed feasible action and child mixtures, and every plane constant is at least (23), then*

$$\underline{V}_t(z, q, b) \leq \mathcal{V}_t(z, q, b) \leq \bar{V}_t(z, q, b) \quad \text{for every } (q, b) \in \mathcal{D}_t(z). \quad (24)$$

At a queried state, an optimizing mixture in (22), followed by its mixed action and child promises, gives a deterministic accepted policy with value at least $\underline{V}_t$. Repeating this procedure using only the current augmented state preserves every continuation cap and the exact root payment. Its loss is at most $\bar{V}_t - \underline{V}_t$.

A uniform certificate requires no unproved smoothness rate. Triangulate the anchor rectangle using stored grid anchors. For a triangle C_0 with vertices i , let

$$\varepsilon_t(z) = \max_{C_0} \min_k \max_{i \in \text{vert}(C_0)} \{H_k(q_i, b_i) - l_i\}. \quad (25)$$

Then $0 \leq \mathcal{V}_t - \underline{V}_t \leq \varepsilon_t(z)$ throughout the rectangle. The order of the minimum and maximum in (25) matters: checking only the minimum-plane gap at each vertex does not in general certify the interior.

Proof. The attainable parent promises are the sum of the action-payment interval and the discounted child intervals, intersected with the parent's cap. This proves (21) by induction. Deterministic convex combinations of two feasible subtree policies preserve the affine payment equalities and all continuation inequalities. Joint concavity of (20), including its inherited-tier argument, proves value concavity.

For a lower anchor, combine the displayed action with the child mixtures. Induction gives the stated reward lower bound. A convex mixture of parent witnesses remains feasible at its mixed inherited tier and promise; concavity bounds its reward from below by the mixed anchor rewards. At the next review, reoptimizing the child mixture cannot reduce that lower bound. This proves the executable lower envelope without storing a separate policy for every history.

For the upper bound, replace each child value by its chosen affine majorant, use $-\ell|x - q| \leq -s(x - q)$, and subtract η times the promise equation. Maximizing the separated action and child-promise terms gives (23). This is weak duality and requires neither an interior promise nor exact optimizer multipliers. The minimum of the resulting planes remains an upper bound.

On a triangle, barycentric interpolation of its anchor rewards is no larger than V_t . For any one plane, the difference from that interpolation is affine and is maximized at a vertex. Minimize this bound over planes, then maximize over triangles. This proves (25). $\square$

The construction compresses *certificates* as well as values. With at most A_0 anchors and H_0 planes per date–regime and at most B_0 successors, dense witness storage is of order $TS_0B_0(A_0^2 + H_0^2)$, not the number of histories. Sparse mixtures reduce storage further. Evaluating a policy still requires a small mixture linear program; building each table requires optimization over child allocations. Neither the anchor count needed for a prescribed tolerance nor the size of rational arithmetic is bounded independently of the problem data. Concave interpolation and stochastic dual cuts are established ideas (Pereira and Pinto, 1991); the contractual conclusion here is that continuous promise interpolation can be accompanied by exact participation feasibility, recursive price witnesses, and an a posteriori policy-loss bound. No such conclusion is inferred from finite-action enumeration alone. In particular, deterministic mixing is invalid for indivisible tiers unless mixtures themselves are admissible actions.

6.2 What is lost by omitting part of the augmented state?

Theorem 6.1 gives zero value to additional history once the displayed state is sufficient. A smaller decision table can still lose value. That loss can be bounded from a certified full-class decision rather than attributed vaguely to “adaptivity.”

Let $\mathcal{A}$ be the affine subspace imposing a predeclared pooling of histories, and let $Y = X \cap \mathcal{A}$ be nonempty. For the general objective in Section 3, define $\text{dist}_Q(x, \mathcal{A}) = \inf_{y \in \mathcal{A}} \|x - y\|_Q$, where $\|u\|_Q^2 = u^\top Q u$.

Proposition 6.3 (Certified value of omitted state). *Let x^* maximize W on X . Then*

$$J - K \geq \frac{1}{2} \text{dist}_Q(x^*, \mathcal{A})^2. \quad (26)$$

For any accepted $\hat{x}$ and valid $U_X \geq J$, put $\epsilon = U_X - W(\hat{x})$. The computable bound

$$J - K \geq \frac{1}{2} \left[\text{dist}_Q(\hat{x}, \mathcal{A}) - \sqrt{2\epsilon} \right]_+^2 \quad (27)$$

is valid. If Q is diagonal with entries $d_n > 0$ and $\mathcal{A}$ only pools decisions in cells C , the squared distance is $\sum_C \sum_{n \in C} d_n (\hat{x}_n - \bar{x}_C)^2$, where $\bar{x}_C = \sum_{n \in C} d_n \hat{x}_n / \sum_{n \in C} d_n$.

Proof. Strong concavity and optimality on the convex set X imply $W(x^*) - W(y) \geq \|x^* - y\|_Q^2 / 2$ for every $y \in X$; the argument also holds at switching kinks by applying strong concavity to the segment from x^* to y and letting its length tend to zero. Minimize the distance over the larger set $\mathcal{A}$. The same inequality gives $\|x^* - \hat{x}\|_Q \leq \sqrt{2\epsilon}$. Distance to a closed affine set is 1-Lipschitz in that norm. The diagonal formula is the weighted least-squares projection. $\square$

Two exact examples isolate the omitted coordinates. In the first, a three-date tree has one root, two equiprobable date-1 regimes, and a common date-2 regime. Let $a = \beta = 1$, $x \in [0, 1]$, stage reward $r_n x - x^2/2$, and reward coefficients $(0, 1/4, -1/4, 0, 0)$ in root, left, right, left-terminal, right-terminal

order. Set root promise 2, date-1 caps $3/4$ and $5/4$, and terminal caps 1. The optimal accepted policy is $(1, 1/2, 1/2, 1/4, 3/4)$, with value $-25/32$. A date-regime table must pool the terminal actions and has optimum $(1, 1/4, 3/4, 1/2, 1/2)$, with value $-27/32$. The gain is $1/16$. The full optimizer reaches the two terminal histories with the same public regime and the same inherited tier $1/2$, but different remaining promises $1/4$ and $3/4$. Thus inherited tier cannot substitute for the promise in these continuation decisions. The curvature lower bound is $1/32$.

Conversely, consider two remaining dates, one regime, the same remaining promise 1, and inherited tiers 0 and 1 occurring with equal probability. With quadratic cost $(x^2 + (1 - x)^2)/2$, incoming switching cost $3|x - q|/2$, and final switching cost $|1 - 2x|/2$, the informed first actions are $1/4$ and $3/4$. Their average value is $-15/16$. A tier-blind common first action is optimally $1/2$, with value -1 . Thus the promise does not replace the inherited tier either. EC Section EC.10 proves both examples, explains the interpolation residual bound, and specifies the independent continuous-state checks.

6.3 A shared-table bridge for restriction values

A restriction on histories is not automatically a restriction that a Bellman recursion may impose independently in each successor. In particular, a time-only table must be the *same table* in all histories. Let a finite public memory m update by $m' = F_t(m, z')$, and let $u = (u_{t,m}) \in [0, 1]^K$ be a table selected once, before uncertainty is resolved. For positive weights $\omega_{t,m}$ and release radius $\delta \geq 0$, impose

$$|x_h - u_{t,m(h)}| \leq \delta \omega_{t,m(h)}. \quad (28)$$

At zero release this is exact history pooling. Eliminating the scalar table entry gives all pairwise inequalities $|x_h - x_{h'}| \leq 2\delta \omega_{t,m}$ within a cell. The factor two matters. Thus this institution is a subclass of Section 4, not a new, incomparable benchmark.

For the remainder of this subsection use the positive-payment scalar Markov institution and the stage reward $g_t(z, q, x) = r_t(z)x - h_t(z)x^2/2 - \ell_t(z)|x - q|$, with $h_t(z) > 0$. Write $\mathcal{V}_t^m(z, q, b; u, \delta)$ for recursion (19) with memory appended and corridor (28) imposed. The global table is a read-only argument of every child problem. The restricted optimum is

$$V_M(\delta) = \max_{u \in [0,1]^K} \mathcal{V}_0^{m_0}(z_0, q_0, b_0; u, \delta). \quad (29)$$

Only tables admitting the specified root promise have finite value. Assume at least one such table exists at zero release.

Theorem 6.4 (Shared-table, continuation-price bridge). *The value in (29) equals the optimized history-tree problem with (28). A maximizing contract has state (t, z, m, q, b) and one precommitted table. The value is nondecreasing, concave, and piecewise quadratic in δ . It need not equal the recursion that reselects the table in each child.*

Moreover, finite families of recursively certified affine planes

$$H_t(q, b; u, \delta) = A_t q + B_t b + \gamma_t^\top u + k_t \delta + C_t$$

give an upper bound on the optimized restricted comparator through the root linear program

$$U_M(\delta) = \max_{u \in [0,1]^K, v} \{v : v \leq H_{0,a}(q_0, b_0; u, \delta) \text{ for every stored } a\}. \quad (30)$$

Every feasible policy with one shared table gives a lower bound $L_M(\delta)$. Consequently,

$$L_M(\delta) - U_M(0) \leq V_M(\delta) - V_M(0) \leq U_M(\delta) - L_M(0). \quad (31)$$

These certificates use state planes, policy witnesses, and a $K+1$ variable master problem, not a query history vector.

For finite horizons the class of recursively generated planes is dual-complete: at any feasible fixed root promise and release, there exists a plane for which maximizing over the table gives $V_M(\delta)$ exactly. This is an existence result, not a polynomial bound on the number of planes needed to discover it. A root plane's release coefficient is the expected discounted sum of its local corridor prices. When a tight plane is also maximized by the optimizing table, this coefficient is a supergradient of V_M ; at a differentiability point it is its marginal release value.

Here is the constructive price recursion. For successor j , choose a valid plane with coefficients $(A_j, B_j, \gamma_j, k_j, C_j)$, or a convex combination of such planes. Put $p_j = P_t(z, j)$, $c = (t, m)$, and let I_j be the unrestricted feasible child-promise interval from Theorem 6.2. Choose payment price $\eta \in \mathbb{R}$, participation price $\chi \geq 0$, corridor prices $\rho^+, \rho^- \geq 0$, and switching tension $|s| \leq \ell_t(z)$. With $a = a_t(z)$, $r = r_t(z)$, $h = h_t(z)$, and $\bar{b} = \bar{b}_t(z)$, define

$$\begin{aligned} d &= r - s - a\eta - \rho^+ + \rho^- + \beta \sum_j p_j A_j, \\ A &= s, \quad B = \eta - \chi, \\ \gamma &= (\rho^+ - \rho^-)e_c + \beta \sum_j p_j \gamma_j, \\ k &= \omega_c(\rho^+ + \rho^-) + \beta \sum_j p_j k_j, \\ C &= \psi_h(d) + \chi \bar{b} + \beta \sum_j p_j \left[C_j + \max_{v \in I_j} (B_j - \eta)v \right], \quad \psi_h(d) = \max_{0 \leq x \leq 1} (dx - hx^2/2). \end{aligned} \quad (32)$$

Terminal coefficients are zero and the terminal promise is zero. An empty unrestricted child domain makes the parent infeasible. The use of unrestricted intervals keeps the plane valid as the shared table changes. Using table-dependent intervals without accounting for their dependence would not establish this affine certificate.

Proof outline. For a fixed table, conditioning and concatenation prove both directions of the state equivalence, without changing the table. Maximizing once over that table proves (29). The scalar interval-intersection criterion eliminates the table and identifies the pairwise release problem. Its value properties follow from Theorem 4.1. For the upper bound, insert the child planes, price the exact payment equation, and add the nonnegative slacks of the participation and two corridor inequalities. The switching inequality is $-\ell|x - q| \leq -s(x - q)$. Maximizing the remaining scalar quadratic and each child promise gives (32); induction and the master problem give (31). To prove completeness, take an optimal tree primal–dual pair, normalize each price by its discounted reach probability, and construct the planes backward. Child payment prices then satisfy $\eta_j - \chi_j = \eta$, every priced slack vanishes, and the local box maximum is attained. Table stationarity makes the optimal shared table maximize the root plane. EC Section EC.12 supplies the complete argument, the normalizations, and the limits of this representation. $\square$

The participation term is essential to this construction: conditional marginal payment prices accumulate along a path as $\eta_j = \eta + \chi_j$. This is the nested-continuation counterpart of $A^\top \mu$ in Proposition 3.1. The coefficient γ keeps the *cross-history* table stationarity, while k keeps the release rent. These two conditions are absent from an unrestricted promise recursion. General signed or nonadditive constraints in (3) do not automatically admit this scalar bridge.

An exact two-period example makes the distinction observable. Set $\beta = 1$, equal child probabilities, all payment coefficients and quadratic curvatures to one, rewards $(1, 0, 2)$, root promise one, and child caps $(2/5, 4/5)$. A shared time-only child tier gives $(x_0, x_L, x_H) = (3/5, 2/5, 2/5)$ and value $37/50$. With no pooling the optimum is $(3/5, 0, 4/5)$ and value $53/50$. Reselecting a separate table in each child falsely reports the latter value even at zero release. Correct release has events at $\delta = 1/10, 3/10, 2/5$; the exact policies, recursively reconstructed prices, and supporting upper planes are independently replayed in EC Section EC.13.

6.4 Choosing an implementable architecture

The hierarchy becomes an endogenous decision when an architecture j has a declared installation and memory cost and a release cost $C_j(\delta)$. For example, $C_j(\delta) = f_j + c_j T|M_j| + a_j \delta + b_j \delta^2/2$, with $b_j \geq 0$, prices a maintained table and the permitted organizational discretion. These are specified design inputs, not estimated causal values of information. The provider chooses

$$\max_{j, 0 \leq \delta \leq \bar{\delta}_j} \{V_{M_j}(\delta) - C_j(\delta)\}.$$

On a regular release segment an interior optimum equates the recursively priced release rent to $C'_j(\delta)$; segment endpoints must also be compared. For a finite offered design menu, bounds L_j, U_j certify the selected feasible design within ε whenever $\max_i (U_i - C_i) - (L_j - C_j) \leq \varepsilon$. Replacing a zero-release contract by full flexibility in the example is worthwhile precisely when its incremental installation cost is below $8/25$. For a linear release cost $C(\delta) = \delta$, the net optimum is $\delta = 1/15$, rather than full flexibility. This decision comparison uses optimized continuation-feasible contracts on both sides.

7 Implemented Gains and Reusable Comparator Certificates

A feasible implemented $\hat{x}$ and a valid upper bound $U_Y \geq K$ give the economic certificate

$$W(\hat{x}) - K \geq W(\hat{x}) - U_Y. \quad (33)$$

An upper bound $U_X \geq J$ separately gives $J - W(\hat{x}) \leq U_X - W(\hat{x})$. The first establishes an improvement over an optimized restriction; the second establishes implementation accuracy. Neither follows from predictor error or an optimizer's termination label. EC Section EC.6 states the box-conjugate bound used by the independent verifier, retaining every participation row, equality, and switching term.

7.1 A comparator envelope requiring no new query solve

Consider a fixed restricted geometry Y and fixed convex cost ϕ with strong-convexity matrix $D_0 \succ 0$:

$$K(h) = \max_{x \in Y} \{(b + U^\top h)^\top x - \phi(x)\}.$$

Let the columns of Z_Y span a linear space containing every difference of two points of Y , and choose

$$L_Y \geq \|UZ_Y(Z_Y^\top D_0 Z_Y)^{-1/2}\|^2.$$

A trace bound on the corresponding positive semidefinite matrix is sufficient and can be evaluated rationally.

Proposition 7.1 (Certified reusable restricted-value envelope). *At an anchor h_i , let $z_i \in Y$ and $U_i \geq K(h_i)$ be certified, and put $\delta_i = U_i - W_{h_i}(z_i)$. Then, for every reward query h ,*

$$K(h) \leq U_i + (U z_i)^\top (h - h_i) + \sqrt{2L_Y \delta_i} \|h - h_i\| + \frac{1}{2} L_Y \|h - h_i\|^2. \quad (34)$$

The minimum of these bounds over a finite cache remains valid. After storing U_i , $U z_i$, δ_i , and h_i , each query uses order r arithmetic operations per anchor, where r is the reward dimension, and no restricted optimization. The bound is valid even when the anchor decision is not exactly optimal.

Proof. Strong convexity gives $\|z_i - x^*(h_i)\|_{D_0}^2 \leq 2\delta_i$. Restricting differences to the displayed linear space bounds the gradient error by $\|U(z_i - x^*)\| \leq \sqrt{2L_Y \delta_i}$. The same strong-monotonicity argument gives an L_Y -Lipschitz gradient $\nabla K(h) = U x^*(h)$. Apply the smooth upper Taylor bound, substitute $U_i \geq K(h_i)$, and bound the anchor-gradient error by Cauchy-Schwarz. EC Section EC.7 gives the full argument, including exact outward square-root rounding. $\square$

This is a nearby-reward reuse mechanism, not a theorem for an arbitrarily changing accepted geometry. Changing continuation matrices, budgets, switching coefficients, or information restrictions requires a new valid cache or a bound that explicitly covers those changes. A loose envelope is still valid but may not certify a positive economic gain. A deployment can then request a new comparator certificate; that solve must be charged. Accepted-policy implementation itself need not perform an economic-comparator batch on every query. Theorem 7.3 supplies a separate valid bound for jointly changing constraints and friction.

7.2 Why the robust outer comparator is locally continuous

Write a designed family as

$$Y_\theta(\rho) = \{x \in C_0 : (A^0 + \rho A_\theta)x \leq b^0 + \rho b_\theta\},$$

where C_0 is the common compact box and equality-restricted class, and θ belongs to a compact uncertainty set. Let η_i bound $\sup_{\theta, x \in C_0} |(A_\theta)_i x - (b_\theta)_i|$ and put $O(\rho) = \{x \in C_0 : A_i^0 x \leq b_i^0 + \rho \eta_i\}$. This radius-dependent outer class contains every $Y_\theta(\rho)$ and equals the nominal class at zero. It is not a fixed large union bound switched on at every positive radius.

Proposition 7.2 (Local outer-comparator loss). *Suppose an anchor in C_0 has slack at least $\gamma_i > 0$ in every perturbed row for $0 \leq \rho \leq \rho_0$. Suppose the objective W_θ is uniformly L_W -Lipschitz on C_0 , whose diameter is D_C . Then*

$$0 \leq \max_{O(\rho)} W_\theta - \max_{Y_\theta(\rho)} W_\theta \leq L_W D_C \max_i \frac{2\rho\eta_i}{\gamma_i + 2\rho\eta_i}. \quad (35)$$

Thus the outer-set component of certificate slack vanishes at least linearly as the radius tends to zero under these assumptions. The constant can be large when an important accepted row has little strict slack.

Proof. An outer-feasible decision violates any true row by at most $2\rho\eta_i$. Mix it with the anchor by the largest fraction in (35). Every resulting true row is feasible, and the common equalities and box remain feasible by convexity. The movement is at most that fraction times D_C . Apply the Lipschitz bound to an outer maximizer. $\square$

The row envelope can be tightened by taking its support over the equality-restricted box rather than an unrestricted box. EC Section EC.8 also gives a valid successive-outer-set refinement. Neither a small uncertainty radius nor set containment alone establishes field robustness: the uncertainty set remains a designed, uncalibrated family. Without a uniform anchor, the linear rate above is not asserted. The finer-radius experiment distinguishes high local sensitivity from an actual positive-radius discontinuity.

7.3 Jointly changing commitments and switching costs

The reward-only envelope does not address the case in which the same operational shock changes revenue, continuation coefficients, capacity, and switching friction. Replacing these correlated constraints by independent row envelopes loses contractual information. We instead retain the actual coefficient vector in a reusable dual certificate.

For this construction, write $Q = H + R^\top R$, where $H \succ 0$ is diagonal, and combine the payment and information equalities into $F_\theta x = f_\theta$. The restricted problem is

$$\begin{aligned} K_\theta &= \max_{x \in Y_\theta} W_\theta(x), & Y_\theta &= \{x \in [\ell, u] : A_\theta x \leq b_\theta, F_\theta x = f_\theta\}, \\ W_\theta(x) &= c_\theta + p_\theta^\top x - \frac{1}{2} x^\top H x - \frac{1}{2} \|Rx\|^2 - \lambda_\theta \sum_j \kappa_j |(Dx - h)_j|. \end{aligned} \quad (36)$$

The common box, H, R, D, h , and positive weights κ are fixed; all displayed θ -dependent coefficients may change together, with $\lambda_\theta \geq 0$. The known constant c_θ permits gain-relative objectives. Each restricted class is nonempty. Define

$$\psi_i(d) = \max_{\ell_i \leq v \leq u_i} \{dv - H_{ii}v^2/2\}, \quad v_i(d) = \min\{u_i, \max\{\ell_i, d/H_{ii}\}\}.$$

A cached price $a = (\alpha, \mu, v, \tau)$ has $\mu \geq 0$ and $|\tau_j| \leq \kappa_j$. Its switching tension at a query is $\lambda_\theta \tau$, not the unscaled tension from the training problem.

Theorem 7.3 (Correlated comparator reuse and its local loss). *For any such cached price, put*

$$d_\theta(a) = p_\theta - R^\top \alpha - A_\theta^\top \mu - F_\theta^\top v - \lambda_\theta D^\top \tau, \quad (37)$$

$$\mathcal{U}_\theta(a) = c_\theta + \frac{1}{2} \|\alpha\|^2 + \mu^\top b_\theta + v^\top f_\theta + \lambda_\theta \tau^\top h + \sum_i \psi_i(d_{\theta,i}(a)). \quad (38)$$

Then $K_\theta \leq \mathcal{U}_\theta(a)$ for every query. The minimum over a finite cache remains valid without a new restricted optimization, an interior feasible anchor, or an unchanged active set.

More precisely, for every feasible restricted z , the certificate gap is the sum of four nonnegative terms:

$$\begin{aligned} \mathcal{U}_\theta(a) - W_\theta(z) &= \frac{1}{2} \|\alpha - Rz\|^2 + \sum_i \left\{ \psi_i(d_{\theta,i}) - d_{\theta,i}z_i + \frac{1}{2} H_{ii}z_i^2 \right\} \\ &\quad + \mu^\top (b_\theta - A_\theta z) + \lambda_\theta \sum_j \left\{ \kappa_j |(Dz - h)_j| - \tau_j (Dz - h)_j \right\}. \end{aligned} \quad (39)$$

Thus the loss separates into resource-price error, coordinate-response error, unused priced continuation capacity, and switching-price error.

For a local statement, let z_0 be the exact restricted optimizer at θ_0 , let the cached price attain equality there, and set $v_0 = d_{\theta_0}(a) - Hz_0 \in N_{[\ell,u]}(z_0)$. Suppose the query optimizer z_θ stays on the priced face: $\mu^\top (b_\theta - A_\theta z_\theta) = 0$, $v_0 \in N_{[\ell,u]}(z_\theta)$, and every switching term in (39) vanishes. With $e_\theta = d_\theta(a) - Hz_\theta - v_0$, one has

$$0 \leq \mathcal{U}_\theta(a) - K_\theta \leq \frac{1}{2} \|R(z_\theta - z_0)\|^2 + \frac{1}{2} e_\theta^\top H^{-1} e_\theta. \quad (40)$$

In particular, locally Lipschitz coefficients and a locally Lipschitz optimizer give a quadratic reuse loss on this priced face. The global upper certificate does not require these local conditions.

Proof. Linearize the resource square with α , lower-bound each absolute value by its normalized tension, and subtract the feasible inequality and equality residuals. Maximizing the remaining separate concave quadratics on the box gives (38). Adding and subtracting their values at z gives (39); equality feasibility removes the equality-price residual. At an exact anchor the resource term implies $\alpha = Rz_0$, and each coordinate is a box maximizer, giving $v_0 \in N_{[\ell,u]}(z_0)$. On the priced face only the first two terms remain. For a box-feasible displacement $y - z_\theta$, the second term is the supremum of $(v_0 + e_\theta)^\top (y - z_\theta) - \|y - z_\theta\|_H^2 / 2$. The normal term is nonpositive; the unrestricted supremum of the remainder is $e_\theta^\top H^{-1} e_\theta / 2$. This proves (40). EC Section EC.11 gives explicit constants and the boundary cases. $\square$

The continuation prices need not be refitted to remain valid. They may, however, become expensive prices for slack that the query optimizer no longer uses. This distinction explains both the benefit of reuse and its failure mode. With affine coefficient dependence in d parameters, cached matrix-price products reduce evaluation to order $n(d + 1)$ arithmetic operations per anchor; exact rational bit cost and all anchor construction remain additional work. This is not the dimension-free reward-only bound of Proposition 7.1, and it is not a measured speedup.

The quadratic conclusion must not be extended across an arbitrary degenerate face. For $|t| \leq 1/4$, the problem $\max\{x - x^2/2 : -1/2 \leq x \leq 1/2, x \leq t, x \leq -t\}$ has the following behavior. An exact price $(1, 0)$ at $t = 0$ gives $\mathcal{U}_t = t$, whereas $K_t = -|t| - t^2/2$. The reuse gap is $2t + t^2/2$ for $t > 0$. The released first constraint has positive cached price and positive slack. Numerical anchor error likewise does not disappear by declaring an anchor exact: (38) remains valid, but the local quadratic assertion requires the exact-anchor assumptions.

Corollary 7.4 (Uniform certification with correlated uncertainty). *Suppose all θ -dependent coefficients in (36) are affine on a compact polytope Θ , and $\lambda_\theta \geq 0$ there. Partition Θ into polytopes C with vertex sets V_C . A fixed implemented policy $\hat{x}$ is feasible for the full accepted class throughout Θ if it satisfies its affine inequalities and equalities at all vertices. For any finite cache $\mathcal{A}$,*

$$\inf_{\theta \in \Theta} \{W_\theta(\hat{x}) - K_\theta\} \geq \min_C \max_{a \in \mathcal{A}} \min_{v \in V_C} \{W_v(\hat{x}) - \mathcal{U}_v(a)\}. \quad (41)$$

Subdivision of cells cannot decrease this lower certificate, and adding cached prices cannot decrease it. Neither operation requires new comparator solves.

Proof. The function $\mathcal{U}_\theta(a)$ is convex in θ : its residuals are affine and each ψ_i is convex. The fixed-policy reward is affine, so their difference is concave and is bounded below on a cell by its minimum vertex value. Choose the best fixed anchor on that cell, then take the worst cell. Convexity also proves the subdivision assertion; retaining old prices proves the cache assertion. $\square$

The anchor must remain fixed inside the vertex minimum in (41). Taking the best anchor separately at vertices and interpolating those best values is generally invalid. The EC gives an exact failure example and a positive-gain example: four cells certify at least 0.04919921 uniformly against an optimized restricted contract under jointly changing reward, friction, and participation coefficients. No sampled parameter is substituted for a continuous uncertainty guarantee. The set is designed, not statistically calibrated.

7.4 A certified cache-governance rule

The global bound in Theorem 7.3 remains valid when its favorable priced face changes. Its quadratic-loss clause is not a license to trust a stale cache. The following rule uses only audited feasible witnesses and the actual query coefficients.

Proposition 7.5 (Quality and refresh certificate). *Let $U = \min_a U_\theta(a)$ be the cached comparator upper bound, and let $z^w \in Y_\theta$ be any independently audited restricted witness. Then*

$$0 \leq U - K_\theta \leq U - W_\theta(z^w).$$

Thus a computed width at most ε certifies cache quality without knowing the query optimizer. Otherwise the controller must refresh the price/witness pair or declare the requested tolerance uncertified. Failure of this sufficient test does not prove that the cache is inaccurate. Deployment additionally requires an audited full-class policy and $W_\theta(\hat{x}) - U$ above the required net improvement.

The width is the exact four-term decomposition in Theorem 7.3 at z^w . A repaired cached table or a feasible state policy from Theorem 6.4 can provide the witness; a repair is not credited as a fresh optimized comparator. No empirical price is interpreted as a directional derivative at degeneracy.

A stronger face test is also available. Store an independent active equality basis, its switching signs, and the exact-anchor prices. At a query, solve only the equality-constrained KKT linear system on that face, then verify *all* primal inequalities, dual signs, switching tensions, and complementarity against the query model. If these checks pass, convex optimality certifies the query restricted optimum; the additional cached-face conditions in Theorem 7.3 then certify its quadratic bound. A nonsingular factorization can be reused only when its coefficient matrix is unchanged. Otherwise factorization and verification are additional costs, not constant-time cache lookups. EC Section EC.14 gives the system and the complete accounting.

7.5 Uniform certificates for a declared adaptive policy

The preceding uniform corollary concerns a fixed policy. To certify an actual adaptive deployment map, partition its parameter domain into simplices and declare $\hat{x}(\theta)$ affine on each simplex. Assume every restricted query set is nonempty, the common box and quadratic/switching geometry stay fixed, and $A_\theta, b_\theta, F_\theta, f_\theta, p_\theta, c_\theta, \lambda_\theta$ are affine, with $\lambda_\theta \geq 0$. Write $E_\theta x = e_\theta$ for the full-class equalities; $F_\theta x = f_\theta$ additionally includes the comparator restriction. The accepted equalities are affine too. The deployed policy is audited against $(A_\theta, b_\theta, E_\theta, e_\theta)$, not against comparator-only pooling rows. Feasibility must hold throughout each cell, not only at sampled queries.

Proposition 7.6 (Adaptive, correlation-preserving cell certificate). *Let the vertices of a simplex be v_i , with policy values $x_i = \hat{x}(v_i)$. A sufficient condition for $A_\theta \hat{x}(\theta) \leq b_\theta$ throughout the simplex is componentwise nonnegativity of all coefficients*

$$B_{ii} = b_i - A_i x_i, \quad B_{ij} = \frac{1}{2}(b_i + b_j - A_i x_j - A_j x_i) \quad (i < j). \quad (42)$$

Apply the same formula to the accepted equalities (E_θ, e_θ) and require every coefficient to be zero; box feasibility is vertex-checkable. Refine cells at switching-sign hyperplanes. For a cached price a , define the quadratic polynomial

$$f_a(\theta) = W_\theta(\hat{x}(\theta)) - \sum_i \lambda_i(\theta) U_{v_i}(a),$$

where the $\lambda_i(\theta)$ in this display are barycentric weights, not friction scales. Its quadratic Bernstein coefficients are

$$g_{ii,a} = f_a(v_i), \quad g_{ij,a} = 2f_a((v_i + v_j)/2) - \frac{1}{2}[f_a(v_i) + f_a(v_j)].$$

If every cell passes the feasibility test, a uniform gain lower bound is

$$\inf_{\theta} \{W_{\theta}(\widehat{x}(\theta)) - K_{\theta}\} \geq \min_{\text{cells}} \max_a \min_{i \leq j} g_{ij,a}. \quad (43)$$

The guarantee includes the interior of every cell and the prescribed boundary policy. Discontinuous piecewise policies require a declared boundary assignment and a certificate for each assigned piece.

The proof expands the correlated products in the nonnegative Bernstein basis. Convexity of $U_{\theta}(a)$ puts it below its vertex chord; the Bernstein minimum bounds the remaining quadratic. Checking only vertices of $A_{\theta}\widehat{x}(\theta)$ is false in general. For example, $A_{\theta} = \theta$, $\widehat{x}(\theta) = 1 - \theta$, $b_{\theta} = 0$ passes both endpoint checks but violates the constraint in the interior. The off-diagonal coefficient in (42) is $-1/2$ and rejects it. EC Section EC.15 proves the result and gives a nonconstant, uniformly feasible policy with certified gain $9/1024$ under jointly changing rewards, a continuation coefficient, and friction scale.

8 Computational Evidence

8.1 Shared restrictions, adaptive certificates, and refresh costs

The new exact two-period checks reconstruct the optimized release policy, its local box maxima, normalized participation prices, shared-table stationarity, and recursive root plane at nine rational releases, including active-cap transitions. Exhaustive rational KKT-basis enumeration independently recovers the tree optimum. All 81 anchor/query release pairs satisfy the supporting-plane inequality. A separate two-plane shared-table master has value $3/8$. The declared adaptive policy in Proposition 7.6 passes its full-cell feasibility test and has uniform gain certificate $9/1024$. Fourteen deliberately invalid witnesses or certificates are rejected; these finite checks supplement, rather than replace, the proofs.

Table 2 varies decision dimension, service count, parameter dimension, cache size, and active-cap instability. The family is an explicit direct sum of two-period accepted-service blocks with shared child tiers, not a claim that a deep nonrecombining tree has been compressed for free. All generated quantities and audits use rational arithmetic. The same exact full-policy generator and its audit are charged to both pipelines. The fresh comparator is a specialized exact solve and meets a stronger zero-gap criterion within the common target; it is not an artificially expensive generic solver.

The strict tolerance triggers refresh on 83 of 84 queries. All post-governance comparator errors are within the declared target, but the cache pipeline is not faster in these measurements. This is useful adverse evidence: global validity alone does not imply that a cached price is economically tight or computationally beneficial. The full phase accounting and isolated-process memory are reported in EC Section EC.17. Historical continuous-state approximation, full-tree comparisons, and unfavorable learning results follow unchanged and are not relabeled as new experiments.

8.2 Continuous-state guarantees without history enumeration

Table 3 reports the inherited continuous-state construction in Section 6.1. Bounds cover every inherited tier and feasible root promise, not only evaluation points. In the four-period design, refining from 25 to 289 anchors per date–regime reduces the worst root bound from 0.113253 to 0.019567. The sixteen-period design stores 800 lower anchors rather than constructing a 65,535-node binary history tree; its coarser grid gives a wider 0.193576 bound. These figures report approximation guarantees and storage, not matched-time superiority or a uniform complexity rate. The six designs, complete rational witnesses, and all eleven negative controls are specified in EC Section EC.10.

8.3 Protocol and common-instance comparator hierarchy

We use two explicitly different families. The first takes the first 24 nominal contexts of the frozen R24 design: 63 history nodes, two service coordinates, six resource–reward coordinates, all continuation rows, and four shared capacities. A pooled action is an equal *amendment* to the existing heterogeneous tier. The fixed-amendment class is the singleton zero amendment because the exact root payment prevents a nonzero uniform change. Calling this an optimized constant actual tier would be incorrect.

The second family has 63 nodes, one tier, a common baseline of $1/2$, an equiprobable binary public regime, and Markov stage rewards, curvature, and conditional continuation caps. All five classes are feasible and pool actual tiers. The exact root commitment fixes the optimized constant tier at $1/2$ but leaves time-only schedules nontrivial. Every adjacent comparison uses the same primitive instance. We vary reward and switching parameters across 24 seeded contexts; EC Section EC.9 gives the integer and rational design. Integer probability weights multiply all conditional weights by 32, so its raw value unit differs from the legacy family. We report shares within each family, not a cardinal comparison between the two families.

Table 4 reports the optimized increments. Full-history amendments retain substantial value beyond two-period memory on the legacy family. In the public-Markov family most of the gain comes from observing the current regime; the additional memory and full-history gains are smaller and can vanish within the numerical bracket on particular instances. This distinction is informative: the total legacy gain is not presented as a uniquely contractual effect or attributed entirely to learning. Regime-only controls omit inherited tiers and promises even in the Markov family. The exact augmented-state result remains Theorem 6.1.

Each optimized class has a rational feasible policy and a separate rational dual upper bound. If these are $[L_j, U_j]$, the certified interval for an increment is $[\max\{0, L_j - U_{j-1}\}, U_j - L_{j-1}]$. We do not interpret a tiny difference between two approximate lower endpoints as a negative true increment. Both the lower-endpoint summaries and these enclosing intervals are retained in the raw record. Exact symbolic tests independently verify the three-node release path, the capacity rent example, and the optimized friction example. Additional rational matrix tests verify the regular-cell release formula. Finite-action promise recursions match exhaustive history enumeration at every attainable root promise; these finite checks supplement the induction proof rather than replace it.

8.4 Radius refinement and comparator reuse

Table 5 recomputes the true and common-outer time-only comparators on eight fixed legacy contexts and ten radii, including much smaller positive radii than the earlier study. Each hidden coefficient vector is held fixed while its radius varies. The outer-set slack tends toward zero and rises steeply before the earlier smallest positive radius. Thus the coarse positive-radius plateau did not establish a discontinuity. The construction already depended continuously on radius; Proposition 7.2 explains its local behavior. This experiment isolates the outer-set contribution and does not conflate it with objective interpolation over uncertainty vertices.

For comparator reuse we fix the legacy time-only geometry and switching coefficient, build four certified anchors, and validate 32 nearby reward queries. The rational trace curvature bound is approximately 0.45830. The largest envelope-minus-certified-lower-value gap is reported in Table 6. The deployed envelope makes no query optimization calls; the 32 validation solves are offline work used to measure tightness. The four initial solves and storage are not free. This is a test of a reusable upper bound and its quality, not a measured end-to-end speedup over the historical robust pipeline.

8.5 Correlated comparator certification without query optimization

Table 7 evaluates Theorem 7.3 on the same joint coefficient family as the outer-comparator study, rather than weakening the accepted constraints. Three historical prices per context form a frozen cache. Its first panel contains the 56 non-anchor queries from the finer-radius record. Its second panel contains 24 new parameter vectors outside those inherited one-dimensional rays. A separate true restricted solve, outer solve, and full accepted solve are used only for offline evaluation at each new query. The deployed cache receives none of their optimizers or values.

Across the 24 new queries, the mean certified upper slack decreases from 0.064069 for the historical outer comparator to 0.016632 for the three-price cache, with both figures rounded upward. All 24 cache bounds are tighter than their matched outer bounds. The implemented full-class policies have positive cached economic-gain lower bounds in every case, with minimum at least 0.762195 in this family’s objective units. Every optimization bracket is below 6×10^{-8} . These are finite, designed comparisons, not a distributional performance or a field-calibration result.

A finite cache is not automatically tight away from its anchors. The one-price column worsens at larger perturbations; adding prices helps without compromising validity. Nor do these calculations establish an end-to-end speed advantage: exact rational arithmetic, all initial solves, the implemented-policy audit, and any requested cache refresh must still be charged. The result is that a new restricted optimization is not logically required to produce the displayed bound. EC Section EC.11.4 specifies all new coefficients, witnesses, source inheritance, and exact checks. Corollary 7.4 separately provides a continuous uncertainty guarantee; the pointwise table is not substituted for that proof.

8.6 Historical learning and timing evidence

The R24 matched study charged optimizer labels, fitting, prediction, auditing, polishing, and fallback against lifted continuation. At the strict target both learned routes fell back in every measured case at 63, 127, and 255 nodes. Their mean total times exceeded the lifted comparator in every configuration, with unfavorable paired intervals. The coarse target also produced no observed finite amortization case. Table 8 retains the strict-target comparisons; the complete phase-resolved record and all less favorable and more favorable individual cases remain in the code/data archive. These are implementation diagnostics, not a second positive contribution about learned optimization.

Let $C_{\text{off}} \geq 0$ be a candidate’s incremental offline cost, and let t_C, t_L be its and the lifted method’s per-query times. The candidate-minus-lifted total cost after Q_0 queries is

$$C_{\text{off}} + Q_0(t_C - t_L) = C_{\text{off}} - Q_0\delta, \quad \delta = t_L - t_C. \quad (44)$$

For positive offline cost a finite positive crossing exists only when $t_C < t_L$, at $Q_0 = C_{\text{off}}/(t_L - t_C)$. Strict savings require a query count above this crossing. The analysis code uses this convention, and the displayed equation now matches it. The historical observations have no positive learned crossing under that protocol.

The historical robust study compared different achieved gains. Its roughly 20-millisecond differences between protected learned policies and robust maximin are therefore descriptive cost accounting, not matched-quality dominance. We do not use these three isolated method points as a computational advantage claim. Their complete gain–cost coordinates remain in the archive. The reward-only envelope applies under explicitly fixed geometry, while Theorem 7.3 retains jointly changing coefficients. Neither is retroactively substituted into the old timings; their certificate quality is assessed separately.

9 Conclusions

An information restriction is a commitment shared across histories, not a collection of independent restrictions that each successor can redesign. The shared-table bridge preserves that commitment in a sufficient-state recursion and in its dual prices. Nested participation prices and globally balanced table prices certify the optimized restricted value, and the resulting release rent supports an explicit net design choice. The earlier frontier and rent results describe its finite-dimensional sensitivity; the state construction identifies when those values can be implemented and certified without a query history vector.

The same distinction governs deployment. A cached upper bound remains valid outside its favorable local face, but its quality requires an audited witness or a refreshed certificate. A parameter-dependent policy requires interior-valid correlated feasibility, not an extrapolation from individually optimized queries. These certificates do not remove the costs of continuous-state approximation, shared design variables, policy generation, or refresh. The exact examples and scaling record make those boundaries observable while retaining the full historical mathematical and computational record.

Data and Code Accessibility

The separate code/data package contains the R25 source, rational certificate records, exact example checks, finite-action recursion checks, replay and build scripts, a point-by-point response, and a preservation manifest. It also contains the complete predecessor manuscript, formal companion, computational record, historical supplement, and earlier repository derivations. None of the predecessor mathematical statements or empirical tables is silently discarded. The current manuscript and its formal electronic companion form the journal-facing argument; the predecessor and computational documents are an indexed research archive rather than additional required theorem layers. No field data are used. Repository delivery is for further author/referee review and is not a submission to the journal.

The R26 directory adds the continuous-state generators, independent rational verifier, point-by-point response to both R24 reports, and byte-identical R25 predecessor copies. The inherited R25/R24 results are replayed rather than relabeled as new measurements. Both formal documents retain their predecessor mathematical statements and proofs; the new continuous-state and omitted-information results are integrated in Section 6.

The R27 directory adds the correlated-price theorem, its uniform-uncertainty extension, 72 new optimization proposals, independent rational checks, a query-only cache interface, and a complete response to both latest reports. It preserves the R26 root files byte for byte and hashes every inherited file. The new evaluations of old radius records are distinguished from new off-ray solves.

The R28 directory adds the shared-table theorem and full proof, endogenous design example, cache-governance and adaptive-policy certificates, exact-arithmetic scaling generator, independent verifier, complete referee response, and byte-preserved R27 reader files. Its current evidence is separate from the inherited R24–R27 observations. The formal proof and arithmetic checks are supplied for further scrutiny, not as an assertion of editorial acceptance.

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Table 1: Closest-result comparison and the role of the shared-table bridge.

Result	Closest established class	Added conclusion and essential structure
Release frontier	Multiparametric QP; Bemporad et al. (2002)	An operational specialization: optimized pooling release with explicit continuation bottlenecks. Positive curvature and polyhedral restrictions are essential; the generic piecewise-affine principle is not new.
Comparator rents	Convex envelopes and sensitivity	Difference of separately optimized continuation rents; an operational comparative static, not a new envelope theorem.
Promise recursion	Recursive promises; constrained MDPs	Exact remaining-payment implementation for this public exit institution. Additivity and sufficient public state are essential.
Shared-table bridge	Nonanticipativity/policy aggregation; Benders decomposition	One common finite-memory table survives every successor; conditional participation prices accumulate along paths; recursive planes certify the optimized pooled class through a table master. Removing the shared table or replacing nested caps by arbitrary rows destroys this conclusion.
Continuous envelopes	Concave interpolation and stochastic dual cuts	Deterministic mixtures preserve actual payment promises and continuation acceptance, with a global a posteriori error bound. Convex feasible graphs are essential.
Correlated cache	Lagrangian weak duality and separable conjugates	Actual correlated continuation coefficients are retained and the four losses are separated. The quadratic rate needs the declared exact-anchor/retained-face conditions; weak duality itself is not claimed as new.
Adaptive cell test	Polynomial Bernstein positivity and convex chords	A continuum certificate for a declared adaptive policy under jointly changing constraint coefficients and friction scale. Fixed quadratic/switching geometry and cellwise affine policies are essential.

Table 2: New exact-arithmetic scaling and cache governance. Each row has 12 queries; the tolerance is 10^{-3} per independent block. “Changes” is the fraction of block active-cap labels changing between successive queries. Total costs include coefficients, the same full-policy generator and audit, witness repair, cache evaluation, and any refresh. Fresh uses an exact specialized restricted solve. Memory is isolated-process peak resident memory; these are synthetic block-separable models, not dense-QP speed comparisons.

n	Services	d	Cache	Changes (%)	Refresh /12	Cached total (ms)	Fresh total (ms)	Memory (MiB)
12	1	1	1	0.0	12	1.90	1.35	16.2
96	2	4	1	0.0	12	15.39	11.07	16.2
96	2	4	4	0.0	11	20.17	11.01	16.3
96	2	4	4	56.8	12	20.94	11.91	16.6
384	4	8	4	55.7	12	81.84	45.58	16.8
384	4	8	8	55.7	12	105.08	44.33	17.7
1536	4	8	8	55.7	12	421.12	177.34	28.3

Table 3: Continuous promised-state certificates on designed Markov models

Dates	Subdivisions	Lower anchors	Upper planes	Uniform gap	Tree nodes
3	2	54	34	0.299587	7
4	4	200	77	0.113253	15
4	8	648	153	0.051939	15
4	16	2,312	322	0.019567	15
8	4	400	175	0.136908	255
16	4	800	362	0.193576	65,535

Lower anchors and upper planes are totals over all dates and both regimes. The uniform gap is the largest certified root loss over both regimes, all inherited tiers, and all feasible promises, rounded upward to six decimals. Tree nodes give the hypothetical full binary history tree, which is not constructed. These rows are not timing comparisons.

Table 4: Value of progressively richer optimized contracts

Expansion	Legacy amendments		Markov service tiers	
	Mean gain	Share (%)	Mean gain	Share (%)
Fixed to time-only	0.116155	7.95	0.127557	0.49
Time-only to regime	0.136312	9.33	26.066328	99.34
Regime to two-period memory	0.224350	15.36	0.028788	0.11
Two-period memory to full history	0.983916	67.36	0.016938	0.06
Total	1.460733	100.00	26.239611	100.00

There are 24 common instances per family and five optimized classes per instance. Means use certified lower endpoints; the complete enclosing increment intervals are in the data. Maximum optimization-bracket widths are $8.13\text{e-}08$ and $2.97\text{e-}07$, respectively. Shares divide each mean increment by the within-family total. Raw value units differ between families. Current regime omits inherited tier and remaining promise.

Table 5: The common outer comparator becomes locally tight

Uncertainty radius	Mean upper bound on outer-set slack
0	0.000000003
1/65536	0.000047365
1/16384	0.000189415
1/4096	0.000757072
1/1024	0.003019002
1/256	0.011836705
1/64	0.045064142
1/16	0.073768497
1/8	0.074890218
1/4	0.078860565

Eight fixed contexts and ten radii, with a directly optimized true and outer comparator at each pair (160 certificates). Entries average the outer certified upper value minus the true certified lower value. At zero the true outer-set slack is exactly zero; the displayed residual is the optimization bracket. Maximum individual bracket width is $3.06\text{e-}08$. This isolates outer-set loss and is not a calibrated uncertainty experiment.

Table 6: Restricted-comparator reuse with fixed geometry

Quantity	Value
Certified anchors	4
Nearby reward queries	32
Restricted optimizations in deployed query evaluation	0
Query optimizations used only for offline validation	32
Curvature trace bound	0.458298401
Mean envelope slack upper bound	0.000129719
Maximum envelope slack upper bound	0.000248488

A common time-only feasible set and switching coefficient are held fixed. Each query evaluates all four analytical anchor envelopes and takes their minimum. The slack is cached upper value minus independently certified query lower value; it therefore encloses the actual upper-bound loss. Anchor construction, policy acceptance checks, storage, and offline validation are separate costs. No end-to-end timing advantage is inferred.

Table 7: Certified comparator upper slack under correlated parameter changes

Radius	Queries	Outer	One price	Three prices
<i>Inherited ray holdouts; no query price reused</i>				
1/65536	8	0.00004737	0.00000001	0.00000001
1/16384	8	0.00018942	0.00000001	0.00000001
1/4096	8	0.00075708	0.00000011	0.00000011
1/1024	8	0.00301901	0.00000169	0.00000169
1/256	8	0.01183671	0.00002694	0.00002694
1/64	8	0.04506415	0.00043085	0.00043085
1/8	8	0.07489022	0.02778576	0.00711323
<i>New off-ray queries; same frozen cache</i>				
1/64	8	0.04463703	0.00073960	0.00073960
1/16	8	0.07337423	0.01428873	0.01428873
1/8	8	0.07419502	0.05643124	0.03486551

Entries are means of $U - L_K$, where L_K is an independently checked restricted feasible value. Each decimal is rounded upward. The one-price cache uses the nominal anchor; the three-price cache also uses the inherited radii 1/16 and 1/4. No anchor query is included in the first panel. Off-ray queries change coefficient directions, not merely radii. Initial prices and the offline validation optimizations are charged separately; these are certificate-quality comparisons, not matched timings.

Table 8: Preserved strict-target end-to-end timing comparisons

Nodes	Lifted (ms)	Tanh (ms)	RBF (ms)
63	10.34	18.84	18.91
127	96.24	112.53	113.62
255	239.93	277.86	274.96

Historical R24 measurements, unchanged. Target 10^{-5} ; 32 held-out contexts and three repetitions per method and size. Both learned routes have 100% fallback at each size. Their paired saving intervals versus lifted continuation are entirely negative. Offline label/fitting costs are retained separately and cannot turn negative online savings into amortization. These are not newly timed R25 measurements.